Chapter 2. Linear Transformations and Matrices

Every linear map is a matrix once bases are chosen

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2.1 Linear transformations, null spaces, and ranges

Motivation. Linear maps preserve the vector operations and are fixed by their values on a basis.

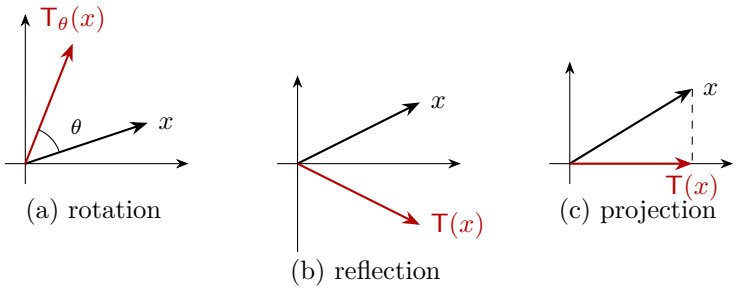
Definition (Linear transformation)

$\mathsf{T}: V \to W$ is **linear** if $\mathsf{T}(x+y) = \mathsf{T}(x) + \mathsf{T}(y)$ and $\mathsf{T}(cx) = c\mathsf{T}(x)$ for all $x, y \in V$, $c \in \mathbb{F}$.

Then $\mathsf{T}(0) = 0$, $\mathsf{T}(x-y) = \mathsf{T}(x) - \mathsf{T}(y)$, $\mathsf{T}(\sum a_i x_i) = \sum a_i \mathsf{T}(x_i)$; linearity $\iff$ $\mathsf{T}(cx+y) = c\mathsf{T}(x) + \mathsf{T}(y)$.

Example 1

$\mathsf{T}(a_1, a_2) = (2a_1 + a_2, a_1)$: for $x = (b_1, b_2)$, $y = (d_1, d_2)$, $\mathsf{T}(cx+y) = (2(cb_1 + d_1) + cb_2 + d_2,\ cb_1 + d_1) = c\mathsf{T}(x) + \mathsf{T}(y)$.



(a) rotation (b) reflection (c) projection

Figure. *Examples 2–4.*

Example 2

Rotation by θ: $r(\cos\alpha, \sin\alpha) \mapsto r(\cos(\alpha+\theta), \sin(\alpha+\theta))$, i.e. $\mathsf{T}_\theta(a_1, a_2) = (a_1\cos\theta - a_2\sin\theta,\ a_1\sin\theta + a_2\cos\theta)$.

Example 3

Reflection about the x-axis: $\mathsf{T}(a_1, a_2) = (a_1, -a_2)$.

Example 4

Projection on the x-axis: $\mathsf{T}(a_1, a_2) = (a_1, 0)$.

Example 5

$\mathsf{T}(A) = A^t$, $\mathrm{M}_{m\times n}(\mathbb{F}) \to \mathrm{M}_{n\times m}(\mathbb{F})$.

Example 6

$\mathsf{T}(f) = f'$, $\mathsf{P}_n(\mathbb{R}) \to \mathsf{P}_{n-1}(\mathbb{R})$: $(ag+h)' = ag' + h'$.

Example 7

$\mathsf{T}(f) = \int_a^b f(t)\,dt$, $C(\mathbb{R}) \to \mathbb{R}$.

The **identity** $\mathsf{I}_V(x) = x$ and **zero transformation** $\mathsf{T}_0(x) = 0$ are linear.

Definition (Null space, range)

For linear $\mathsf{T} : V \to W$, the **null space** (kernel) is $\mathrm{N}(\mathsf{T}) = \{x \in V : \mathsf{T}(x) = 0\}$ and the **range** (image) is $\mathrm{R}(\mathsf{T}) = \{\mathsf{T}(x) : x \in V\}$.

Example 8

$\mathrm{N}(\mathsf{I}) = \{0\}$, $\mathrm{R}(\mathsf{I}) = V$; $\mathrm{N}(\mathsf{T}_0) = V$, $\mathrm{R}(\mathsf{T}_0) = \{0\}$.

Example 9

For $\mathsf{T}(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$ on $\mathbb{R}^3$: $\mathrm{N}(\mathsf{T}) = \{(a, a, 0) : a \in \mathbb{R}\}$ and $\mathrm{R}(\mathsf{T}) = \mathbb{R}^2$.

Theorem 2.1

For linear $\mathsf{T} : V \to W$, $\mathrm{N}(\mathsf{T})$ and $\mathrm{R}(\mathsf{T})$ are subspaces of V and W.

Proof. $\mathsf{T}(0) = 0$ puts 0 in both. For $x, y \in \mathrm{N}(\mathsf{T})$: $\mathsf{T}(x + y) = \mathsf{T}(cx) = 0$. In $\mathrm{R}(\mathsf{T})$: $\mathsf{T}(x) + \mathsf{T}(y) = \mathsf{T}(x + y)$, $c\mathsf{T}(x) = \mathsf{T}(cx)$. Apply Theorem 1.3. ■

Theorem 2.2

If $\beta = \{v_1, \ldots, v_n\}$ is a basis for V and $\mathsf{T} : V \to W$ is linear, then $\mathrm{R}(\mathsf{T}) = \mathrm{span}(\mathsf{T}(\beta)) = \mathrm{span}\{\mathsf{T}(v_1), \ldots, \mathsf{T}(v_n)\}$.

Proof. $\mathrm{span}(\mathsf{T}(\beta)) \subseteq \mathrm{R}(\mathsf{T})$ by Theorem 1.5; conversely $\mathsf{T}(\sum a_i v_i) = \sum a_i \mathsf{T}(v_i)$. ■

(Also for infinite bases.)

Example 10

$\mathsf{T} : \mathsf{P}_2(\mathbb{R}) \to \mathrm{M}_{2\times 2}(\mathbb{R})$, $\mathsf{T}(f) = \begin{pmatrix} f(1)-f(2) & 0 \\ 0 & f(0) \end{pmatrix}$: $\mathrm{R}(\mathsf{T}) = \mathrm{span}\{\mathsf{T}(1), \mathsf{T}(x), \mathsf{T}(x^2)\} = \mathrm{span}\left\{\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} -3 & 0 \\ 0 & 0 \end{pmatrix}\right\}$, of dimension 2.

Definition (Nullity, rank)

If $\mathrm{N}(\mathsf{T})$, $\mathrm{R}(\mathsf{T})$ are finite-dimensional, $\mathrm{nullity}(\mathsf{T}) = \dim \mathrm{N}(\mathsf{T})$ and $\mathrm{rank}(\mathsf{T}) = \dim \mathrm{R}(\mathsf{T})$.

Theorem 2.3 (Dimension theorem)

If $\mathsf{T} : V \to W$ is linear and V is finite-dimensional, then $\mathrm{nullity}(\mathsf{T}) + \mathrm{rank}(\mathsf{T}) = \dim(V)$.

Proof. Extend a basis $\{v_1, \ldots, v_k\}$ of $\mathrm{N}(\mathsf{T})$ to a basis $\{v_1, \ldots, v_n\}$ of V (Corollary); we show $S = \{\mathsf{T}(v_{k+1}), \ldots, \mathsf{T}(v_n)\}$ is a basis of $\mathrm{R}(\mathsf{T})$ with $n - k$ distinct elements.

Step 1 (spans). $\mathrm{R}(\mathsf{T}) = \mathrm{span}\{\mathsf{T}(v_1), \ldots, \mathsf{T}(v_n)\} = \mathrm{span}(S)$ (Theorem 2.2; $\mathsf{T}(v_i) = 0$ for $i \le k$).

Step 2 (independent). $\sum_{i>k} b_i \mathsf{T}(v_i) = 0 \Rightarrow \sum_{i>k} b_i v_i \in \mathrm{N}(\mathsf{T}) \Rightarrow \sum_{i>k} b_i v_i = \sum_{i \le k} c_i v_i \Rightarrow$ all $b_i = 0$ (also giving distinctness). ■

Theorem 2.4

Linear T is one-to-one iff $\mathrm{N}(\mathsf{T}) = \{0\}$.

Proof. One-to-one: $\mathsf{T}(x) = 0 = \mathsf{T}(0) \Rightarrow x = 0$. Conversely $\mathsf{T}(x) = \mathsf{T}(y) \Rightarrow \mathsf{T}(x - y) = 0 \Rightarrow x = y$. ■

Theorem 2.5

If $\dim V = \dim W < \infty$ and $\mathsf{T} : V \to W$ is linear, the following are equivalent: (a) T is one-to-one;
(b) T is onto;
(c) $\mathrm{rank}(\mathsf{T}) = \dim(V)$.

Proof. By Theorems 2.3 and 2.4: T one-to-one $\iff$ $\mathrm{N}(\mathsf{T}) = \{0\}$ $\iff$ $\mathrm{nullity}(\mathsf{T}) = 0$ $\iff$ $\mathrm{rank}(\mathsf{T}) = \dim V = \dim W$ $\iff$ $\mathrm{R}(\mathsf{T}) = W$, the last step by Theorem 1.11. ■

Finite dimension and linearity are both essential.

Example 11

$\mathsf{T} : \mathsf{P}_2(\mathbb{R}) \to \mathsf{P}_3(\mathbb{R})$, $\mathsf{T}(f) = 2f' + \int_0^x 3f(t)\,dt$: $\mathrm{R}(\mathsf{T}) = \mathrm{span}\{3x,\ 2 + \frac{3}{2}x^2,\ 4x + x^3\}$ has rank $3 < 4$: not onto; nullity $3 - 3 = 0$: one-to-one.

Example 12

$\mathsf{T}(a_1, a_2) = (a_1 + a_2, a_1)$ on $\mathbb{F}^2$: $\mathrm{N}(\mathsf{T}) = \{0\}$, so onto (Theorem 2.5).

A one-to-one linear T maps S to an independent set iff S is independent.

Example 13

With one-to-one $\mathsf{T}(a_0 + a_1 x + a_2 x^2) = (a_0, a_1, a_2)$, $\{2 - x + 3x^2,\ x + x^2,\ 1 - 2x^2\}$ is independent since $\{(2, -1, 3), (0, 1, 1), (1, 0, -2)\}$ is.

Theorem 2.6

Let $\{v_1, \ldots, v_n\}$ be a basis for V. For any $w_1, \ldots, w_n \in W$ there is exactly one linear $\mathsf{T} : V \to W$ with $\mathsf{T}(v_i) = w_i$.

Proof. For $x = \sum a_i v_i$ (unique coefficients, Theorem 1.8) set $\mathsf{T}(x) = \sum a_i w_i$.

Step 1 (linear). If $y = \sum b_i v_i$, then $cx + y = \sum (ca_i + b_i) v_i$, so $\mathsf{T}(cx + y) = \sum (ca_i + b_i) w_i = c\mathsf{T}(x) + \mathsf{T}(y)$.

Step 2 (values). $\mathsf{T}(v_i) = w_i$.

Step 3 (unique). A linear U with $\mathsf{U}(v_i) = w_i$ satisfies $\mathsf{U}(x) = \sum a_i \mathsf{U}(v_i) = \sum a_i w_i = \mathsf{T}(x)$. ■

Corollary

If $\{v_1, \ldots, v_n\}$ is a basis for V and linear $\mathsf{U}, \mathsf{T} : V \to W$ agree on each v_i, then $\mathsf{U} = \mathsf{T}$.

Proof. Uniqueness in Theorem 2.6. ■

Example 14

Linear U with $\mathsf{U}(1,2) = (3,3)$, $\mathsf{U}(1,1) = (1,3)$ equals $\mathsf{T}(a_1, a_2) = (2a_2 - a_1, 3a_1)$, which agrees on the basis $\{(1,2),(1,1)\}$.

2.2 The matrix representation of a linear transformation

Motivation. Ordered bases turn vectors into columns and linear maps into matrices.

Definition (Ordered basis; coordinate vector)

An **ordered basis** is a basis with a fixed order. For $\beta = \{u_1, \ldots, u_n\}$ and $x = \sum a_i u_i$, the **coordinate vector** is $[x]_\beta = (a_1, \ldots, a_n)^t \in \mathbb{F}^n$. The **standard ordered bases** are $\{e_1, \ldots, e_n\}$ of $\mathbb{F}^n$ and $\{1, x, \ldots, x^n\}$ of $\mathsf{P}_n(\mathbb{F})$.

Example 1

$\{e_1, e_2, e_3\} \neq \{e_2, e_1, e_3\}$ as ordered bases of $\mathbb{F}^3$.

Example 2

$f = 4 + 6x - 7x^2$, standard β of $\mathsf{P}_2(\mathbb{R})$: $[f]_\beta = (4, 6, -7)^t$.

Definition (Matrix representation)

Let $\beta = \{v_1, \ldots, v_n\}$, $\gamma = \{w_1, \ldots, w_m\}$ be ordered bases of V, W and $\mathsf{T} : V \to W$ linear, with $\mathsf{T}(v_j) = \sum_{i=1}^m a_{ij} w_i$. The $m \times n$ matrix $A = (a_{ij}) = [\mathsf{T}]_\beta^\gamma$ is the **matrix representation** of T; its j-th column is $[\mathsf{T}(v_j)]_\gamma$. If $V = W$, $\beta = \gamma$, write $[\mathsf{T}]_\beta$.

By the Corollary to Theorem 2.6, $[\mathsf{U}]_\beta^\gamma = [\mathsf{T}]_\beta^\gamma$ implies $\mathsf{U} = \mathsf{T}$.

Example 3

$\mathsf{T}(a_1, a_2) = (a_1 + 3a_2, 0, 2a_1 - 4a_2)$, standard bases: $\mathsf{T}(e_1) = (1, 0, 2)$, $\mathsf{T}(e_2) = (3, 0, -4)$, $[\mathsf{T}]_\beta^\gamma = \begin{pmatrix} 1 & 3 \\ 0 & 0 \\ 2 & -4 \end{pmatrix}$. With $\gamma' = \{e_3, e_2, e_1\}$ the rows are reversed.

Example 4

$\mathsf{T}(f) = f'$, $\mathsf{P}_3(\mathbb{R}) \to \mathsf{P}_2(\mathbb{R})$, standard bases: $\mathsf{T}(1) = 0$, $\mathsf{T}(x) = 1$, $\mathsf{T}(x^2) = 2x$, $\mathsf{T}(x^3) = 3x^2$, $[\mathsf{T}]_\beta^\gamma = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$.

Definition (Sum and scalar multiple of maps)

$(\mathsf{T}+\mathsf{U})(x) = \mathsf{T}(x)+\mathsf{U}(x)$ and $(a\mathsf{T})(x) = a\mathsf{T}(x)$.

Theorem 2.7

Let $\mathsf{T}, \mathsf{U} : V \to W$ be linear.
(a) $a\mathsf{T}+\mathsf{U}$ is linear for all $a \in \mathbb{F}$.
(b) With these operations, the linear maps $V \to W$ form a vector space over $\mathbb{F}$.

Proof.

(a) $(a\mathsf{T}+\mathsf{U})(cx+y) = a\mathsf{T}(cx+y)+\mathsf{U}(cx+y) = c(a\mathsf{T}+\mathsf{U})(x)+(a\mathsf{T}+\mathsf{U})(y)$.

(b) By (a) the linear maps form a subspace of $\mathcal{F}(V,W)$ (it contains T_0). ■

$\mathcal{L}(V,W)$ denotes this space, and $\mathcal{L}(V) = \mathcal{L}(V,V)$.

Theorem 2.8

For finite-dimensional V, W with ordered bases β, γ and linear $\mathsf{T}, \mathsf{U} : V \to W$:
(a) $[\mathsf{T}+\mathsf{U}]_\beta^\gamma = [\mathsf{T}]_\beta^\gamma + [\mathsf{U}]_\beta^\gamma$;
(b) $[a\mathsf{T}]_\beta^\gamma = a[\mathsf{T}]_\beta^\gamma$.

Proof. If $\mathsf{T}(v_j) = \sum_i a_{ij}w_i$ and $\mathsf{U}(v_j) = \sum_i b_{ij}w_i$, then $(\mathsf{T}+\mathsf{U})(v_j) = \sum_i (a_{ij}+b_{ij})w_i$ and $(a\mathsf{T})(v_j) = \sum_i aa_{ij}w_i$. ■

Example 5

$\mathsf{T}(a_1,a_2) = (a_1+3a_2, 0, 2a_1-4a_2)$, $\mathsf{U}(a_1,a_2) = (a_1-a_2, 2a_1, 3a_1+2a_2)$: $(\mathsf{T}+\mathsf{U})(a_1,a_2) = (2a_1+2a_2, 2a_1, 5a_1-2a_2)$ and $\begin{pmatrix} 2 & 2 \\ 2 & 0 \\ 5 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 0 & 0 \\ 2 & -4 \end{pmatrix} + \begin{pmatrix} 1 & -1 \\ 2 & 0 \\ 3 & 2 \end{pmatrix}$.

2.3 Composition of linear transformations and matrix multiplication

Motivation. Matrix multiplication is defined so that it matches composition of maps.

Theorem 2.9

If $\mathsf{T} : V \to W$ and $\mathsf{U} : W \to Z$ are linear, so is $\mathsf{UT} : V \to Z$.

Proof. $\mathsf{UT}(ax+y) = \mathsf{U}(a\mathsf{T}(x)+\mathsf{T}(y)) = a\mathsf{UT}(x)+\mathsf{UT}(y)$. ■

Theorem 2.10

For $\mathsf{T}, \mathsf{U}_1, \mathsf{U}_2 \in \mathcal{L}(V)$:
(a) $\mathsf{T}(\mathsf{U}_1+\mathsf{U}_2) = \mathsf{TU}_1+\mathsf{TU}_2$ and $(\mathsf{U}_1+\mathsf{U}_2)\mathsf{T} = \mathsf{U}_1\mathsf{T}+\mathsf{U}_2\mathsf{T}$;
(b) $\mathsf{T}(\mathsf{U}_1\mathsf{U}_2) = (\mathsf{TU}_1)\mathsf{U}_2$;
(c) $\mathsf{TI} = \mathsf{IT} = \mathsf{T}$;
(d) $a(\mathsf{U}_1\mathsf{U}_2) = (a\mathsf{U}_1)\mathsf{U}_2 = \mathsf{U}_1(a\mathsf{U}_2)$.

Proof [Supplement]. Evaluate at x. (a) $\mathsf{T}(\mathsf{U}_1x+\mathsf{U}_2x) = \mathsf{TU}_1x+\mathsf{TU}_2x$ (linearity); $(\mathsf{U}_1+\mathsf{U}_2)(\mathsf{T}x) = \mathsf{U}_1\mathsf{T}x+\mathsf{U}_2\mathsf{T}x$.

(b) Composition is associative.

(c) Clear.

(d) $a\mathsf{U}_1(\mathsf{U}_2x) = (a\mathsf{U}_1)(\mathsf{U}_2x) = \mathsf{U}_1(a\mathsf{U}_2x)$ by linearity of U_1. ■

Let $\mathsf{T}: V \to W$, $\mathsf{U}: W \to Z$ with ordered bases $\alpha = \{v_j\}$, $\beta = \{w_k\}$, $\gamma = \{z_i\}$, $A = [\mathsf{U}]_\beta^\gamma$, $B = [\mathsf{T}]_\alpha^\beta$. Then

$$\mathsf{UT}(v_j) = \mathsf{U}\Big(\sum_k B_{kj} w_k\Big) = \sum_k B_{kj} \sum_i A_{ik} z_i = \sum_i \Big(\sum_k A_{ik} B_{kj}\Big) z_i.$$

Definition (Matrix product)

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$ and $B \in \mathrm{M}_{n\times p}(\mathbb{F})$, $AB \in \mathrm{M}_{m\times p}(\mathbb{F})$ has $(AB)_{ij} = \sum_{k=1}^n A_{ik}B_{kj}$.

Example 1

$\begin{pmatrix}1&2&1\\0&4&-1\end{pmatrix}\begin{pmatrix}4\\2\\5\end{pmatrix} = \begin{pmatrix}13\\3\end{pmatrix}$. Not commutative: $\begin{pmatrix}1&1\\0&0\end{pmatrix}\begin{pmatrix}0&1\\1&0\end{pmatrix} = \begin{pmatrix}1&1\\0&0\end{pmatrix}$ but $\begin{pmatrix}0&1\\1&0\end{pmatrix}\begin{pmatrix}1&1\\0&0\end{pmatrix} = \begin{pmatrix}0&0\\1&1\end{pmatrix}$. Also $(AB)^t = B^tA^t$, since both have (i,j) entry $\sum_k A_{jk}B_{ki}$.

Theorem 2.11

With α, β, γ ordered bases of finite-dimensional V, W, Z and linear $\mathsf{T}: V \to W$, $\mathsf{U}: W \to Z$: $[\mathsf{UT}]_\alpha^\gamma = [\mathsf{U}]_\beta^\gamma [\mathsf{T}]_\alpha^\beta$.

Proof. By the computation above, $([\mathsf{UT}]_\alpha^\gamma)_{ij} = \sum_k A_{ik}B_{kj}$. ■

Corollary

For $\mathsf{T}, \mathsf{U} \in \mathcal{L}(V)$ and an ordered basis β, $[\mathsf{UT}]_\beta = [\mathsf{U}]_\beta[\mathsf{T}]_\beta$.

Proof. Theorem 2.11 with $\alpha = \beta = \gamma$. ■

Example 2

$\mathsf{U}(f) = f'$ ($\mathsf{P}_3 \to \mathsf{P}_2$) and $\mathsf{T}(f) = \int_0^x f(t)\,dt$ ($\mathsf{P}_2 \to \mathsf{P}_3$) give $\mathsf{UT} = \mathsf{I}$:

$$[\mathsf{U}][\mathsf{T}] = \begin{pmatrix}0&1&0&0\\0&0&2&0\\0&0&0&3\end{pmatrix}\begin{pmatrix}0&0&0\\1&0&0\\0&\frac12&0\\0&0&\frac13\end{pmatrix} = I_3.$$

Definition (Kronecker delta; identity matrix)

$\delta_{ij} = 1$ if $i = j$ and 0 otherwise; the $n \times n$ **identity matrix** is $(I_n)_{ij} = \delta_{ij}$.

Theorem 2.12

Let A be $m \times n$, B, C be $n \times p$, D, E be $q \times m$. Then
(a) $A(B + C) = AB + AC$ and $(D + E)A = DA + EA$;
(b) $a(AB) = (aA)B = A(aB)$;
(c) $I_mA = A = AI_n$;
(d) $[\mathsf{I}_V]_\beta = I_n$ for an n-dimensional V with ordered basis β.

Proof.

(a) $(A(B+C))_{ij} = \sum_k A_{ik}(B_{kj}+C_{kj}) = (AB)_{ij} + (AC)_{ij}$; similarly for $(D+E)A$.

(b) [Supplement] All three have entries $a\sum_k A_{ik}B_{kj}$.

(c) $(I_m A)_{ij} = \sum_k \delta_{ik}A_{kj} = A_{ij}$, and similarly for AI_n.

(d) [Supplement] $\mathsf{I}_V(v_j) = v_j$, whose coordinate vector is e_j. ■

Corollary

$A(\sum_{i=1}^k a_iB_i) = \sum_i a_iAB_i$ and $(\sum_i a_iC_i)A = \sum_i a_iC_iA$.

Proof. Induction on k using Theorem 2.12(a), (b). ■

Powers: $A^0 = I$, $A^k = A^{k-1}A$. No cancellation: $A = \left(\begin{smallmatrix}0&0\\1&0\end{smallmatrix}\right) \neq O$ has $A^2 = O$.

Theorem 2.13

Let A be $m \times n$, B be $n \times p$, and u_j, v_j the j-th columns of AB and B. Then (a) $u_j = Av_j$; (b) $v_j = Be_j$.

Proof.

(a) $(u_j)_i = (AB)_{ij} = \sum_k A_{ik}B_{kj} = (Av_j)_i$.

(b) (a) for $BI_p = B$: column j is Be_j. ■

Likewise row i of AB is (row i of A)$\cdot B$.

Theorem 2.14

For linear $\mathsf{T}: V \to W$ with ordered bases β, γ and $u \in V$: $[\mathsf{T}(u)]_\gamma = [\mathsf{T}]_\beta^\gamma [u]_\beta$.

Proof. Let $f(a) = au$ $(\mathbb{F} \to V)$, $g(a) = a\mathsf{T}(u)$ $(\mathbb{F} \to W)$, $\alpha = \{1\}$. Then $g = \mathsf{T}f$, $[f]_\alpha^\beta = [u]_\beta$, $[g]_\alpha^\gamma = [\mathsf{T}(u)]_\gamma$, and by Theorem 2.11 $[\mathsf{T}(u)]_\gamma = [\mathsf{T}f]_\alpha^\gamma = [\mathsf{T}]_\beta^\gamma [f]_\alpha^\beta = [\mathsf{T}]_\beta^\gamma [u]_\beta$. ■

Example 3

$\mathsf{T}(f) = f'$, $p = 2 - 4x + x^2 + 3x^3$: $[\mathsf{T}]_\beta^\gamma (2,-4,1,3)^t = (-4,2,9)^t = [p']_\gamma$.

Definition (Left-multiplication transformation)

For $A \in \mathrm{M}_{m\times n}(\mathbb{F})$, $\mathsf{L}_A: \mathbb{F}^n \to \mathbb{F}^m$ is $\mathsf{L}_A(x) = Ax$.

Example 4

For $A = \left(\begin{smallmatrix}1&2&1\\0&1&2\end{smallmatrix}\right)$, $\mathsf{L}_A(1,3,-1)^t = (6,1)^t$.

Theorem 2.15

L_A is linear. For $B \in \mathrm{M}_{m\times n}(\mathbb{F})$ and standard ordered bases β, γ of $\mathbb{F}^n, \mathbb{F}^m$:
(a) $[\mathsf{L}_A]_\beta^\gamma = A$;
(b) $\mathsf{L}_A = \mathsf{L}_B$ iff $A = B$;

(c) $\mathsf{L}_{A+B} = \mathsf{L}_A + \mathsf{L}_B$ and $\mathsf{L}_{aA} = a\mathsf{L}_A$;
(d) if $\mathsf{T} : \mathbb{F}^n \to \mathbb{F}^m$ is linear, there is a unique C with $\mathsf{T} = \mathsf{L}_C$, namely $C = [\mathsf{T}]_\beta^\gamma$;
(e) if E is $n \times p$, then $\mathsf{L}_{AE} = \mathsf{L}_A\mathsf{L}_E$;
(f) if $m = n$, then $\mathsf{L}_{I_n} = \mathsf{I}_{\mathbb{F}^n}$.

Proof. Linearity follows from Theorem 2.12.

(a) Column j of $[\mathsf{L}_A]_\beta^\gamma$ is Ae_j, column j of A (Theorem 2.13).

(b) If $\mathsf{L}_A = \mathsf{L}_B$, then $A = [\mathsf{L}_A]_\beta^\gamma = [\mathsf{L}_B]_\beta^\gamma = B$ by (a).

(c) [Supplement] $(A+B)x = Ax + Bx$ and $(aA)x = a(Ax)$.

(d) Put $C = [\mathsf{T}]_\beta^\gamma$. By Theorem 2.14, $\mathsf{T}(x) = [\mathsf{T}(x)]_\gamma = C[x]_\beta = Cx = \mathsf{L}_C(x)$. Uniqueness is (b).

(e) $(AE)e_j = A(Ee_j)$ (Theorem 2.13): L_{AE}, $\mathsf{L}_A\mathsf{L}_E$ agree on a basis (Corollary to Theorem 2.6).

(f) [Supplement] $I_n x = x$. ■

Theorem 2.16

If $A(BC)$ is defined, so is $(AB)C$, and $A(BC) = (AB)C$.

Proof. By Theorem 2.15(e) and associativity of composition, $\mathsf{L}_{A(BC)} = \mathsf{L}_A\mathsf{L}_{BC} = \mathsf{L}_A(\mathsf{L}_B\mathsf{L}_C) = (\mathsf{L}_A\mathsf{L}_B)\mathsf{L}_C = \mathsf{L}_{(AB)C}$; apply (b). ■

Incidence matrices. For a relation on $\{1, \ldots, n\}$, $A_{ij} = 1$ if $i \to j$, else 0; $(A^k)_{ij}$ counts k-step paths $i \to \cdots \to j$. For a **dominance relation** (exactly one of $i \to j$, $j \to i$ for $i \neq j$; $A_{ii} = 0$), row sums of $A + A^2$ rank the members.

2.4 Invertibility and isomorphisms

Motivation. Finite-dimensional spaces over $\mathbb{F}$ are classified up to isomorphism by dimension.

Definition (Inverse)

$\mathsf{U} : W \to V$ is an **inverse** of linear $\mathsf{T} : V \to W$ if $\mathsf{TU} = \mathsf{I}_W$ and $\mathsf{UT} = \mathsf{I}_V$; T is then **invertible**, the inverse is unique and denoted T^{-1}.

$(\mathsf{TU})^{-1} = \mathsf{U}^{-1}\mathsf{T}^{-1}$, $(\mathsf{T}^{-1})^{-1} = \mathsf{T}$; invertible $\iff$ one-to-one and onto, so for $\dim V = \dim W < \infty$: invertible $\iff$ $\operatorname{rank}(\mathsf{T}) = \dim V$ (Theorem 2.5).

Example 1

$\mathsf{T}(a + bx) = (a, a + b)$ from $\mathsf{P}_1(\mathbb{R})$ to $\mathbb{R}^2$ has $\mathsf{T}^{-1}(c, d) = c + (d - c)x$.

Theorem 2.17

If $\mathsf{T} : V \to W$ is linear and invertible, then T^{-1} is linear.

Proof. With $x_i = \mathsf{T}^{-1}(y_i)$, $\mathsf{T}^{-1}(cy_1 + y_2) = \mathsf{T}^{-1}(c\mathsf{T}(x_1) + \mathsf{T}(x_2)) = \mathsf{T}^{-1}(\mathsf{T}(cx_1 + x_2)) = cx_1 + x_2 = c\mathsf{T}^{-1}(y_1) + \mathsf{T}^{-1}(y_2)$. ■

Definition (Invertible matrix)

$A \in \mathrm{M}_{n\times n}(\mathbb{F})$ is **invertible** if $AB = BA = I$ for some B. Then B is unique ($C = CI = C(AB) = (CA)B = B$), the **inverse** A^{-1}.

Example 2

$\left(\begin{smallmatrix} 5 & 7 \\ 2 & 3 \end{smallmatrix}\right)^{-1} = \left(\begin{smallmatrix} 3 & -7 \\ -2 & 5 \end{smallmatrix}\right)$.

Lemma

Let $\mathsf{T} : V \to W$ be an invertible linear map. Then V is finite-dimensional iff W is, and then $\dim V = \dim W$.

Proof. A finite basis β of V gives $W = \mathrm{R}(\mathsf{T}) = \operatorname{span}\mathsf{T}(\beta)$ (Theorem 2.2); conversely use T^{-1}. Then $\operatorname{nullity}\mathsf{T} = 0$, $\operatorname{rank}\mathsf{T} = \dim W$, so $\dim V = \dim W$ (Theorem 2.3). ■

Theorem 2.18

Let V, W be finite-dimensional with ordered bases β, γ and $\mathsf{T} : V \to W$ linear. Then T is invertible iff $[\mathsf{T}]_\beta^\gamma$ is invertible, and then $[\mathsf{T}^{-1}]_\gamma^\beta = ([\mathsf{T}]_\beta^\gamma)^{-1}$.

Proof.

Step 1 (only if). $\dim V = \dim W = n$ (Lemma). By Theorems 2.11, 2.12(d), $I_n = [\mathsf{I}_V]_\beta = [\mathsf{T}^{-1}\mathsf{T}]_\beta = [\mathsf{T}^{-1}]_\gamma^\beta[\mathsf{T}]_\beta^\gamma$, and likewise $[\mathsf{T}]_\beta^\gamma[\mathsf{T}^{-1}]_\gamma^\beta = I_n$.

Step 2 (if). $A = [\mathsf{T}]_\beta^\gamma$ has inverse B, so A is square: $\dim V = \dim W = n$. Theorem 2.6 gives linear U with $\mathsf{U}(w_j) = \sum_i B_{ij}v_i$, i.e. $[\mathsf{U}]_\gamma^\beta = B$. Then $[\mathsf{UT}]_\beta = BA = I_n = [\mathsf{I}_V]_\beta$, so $\mathsf{UT} = \mathsf{I}_V$; likewise $\mathsf{TU} = \mathsf{I}_W$. ■

Example 3

Example 1, standard bases: $[\mathsf{T}] = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, $[\mathsf{T}^{-1}] = \left(\begin{smallmatrix} 1 & 0 \\ -1 & 1 \end{smallmatrix}\right)$, mutually inverse.

Corollary 1

For $\mathsf{T} \in \mathcal{L}(V)$, V finite-dimensional with ordered basis β: T is invertible iff $[\mathsf{T}]_\beta$ is, and $[\mathsf{T}^{-1}]_\beta = ([\mathsf{T}]_\beta)^{-1}$.

Proof. Theorem 2.18 with $W = V$, $\gamma = \beta$. ■

Corollary 2

$A \in \mathrm{M}_{n\times n}(\mathbb{F})$ is invertible iff L_A is, and $(\mathsf{L}_A)^{-1} = \mathsf{L}_{A^{-1}}$.

Proof. Theorem 2.18 with standard bases and Theorem 2.15(a), (d): $[(\mathsf{L}_A)^{-1}] = A^{-1}$, so $(\mathsf{L}_A)^{-1} = \mathsf{L}_{A^{-1}}$. ■

Definition (Isomorphism)

V is **isomorphic** to W ($V \cong W$) if there is an invertible linear $\mathsf{T} : V \to W$, an **isomorphism**. "Is isomorphic to" is an equivalence relation.

Example 4

$\mathsf{T}(a_1, a_2) = a_1 + a_2 x$ is an isomorphism $\mathbb{F}^2 \to \mathsf{P}_1(\mathbb{F})$.

Example 5

$\mathsf{T}(f) = \begin{pmatrix} f(1) & f(2) \\ f(3) & f(4) \end{pmatrix}$ from $\mathsf{P}_3(\mathbb{R})$ to $\mathrm{M}_{2\times 2}(\mathbb{R})$ is linear, $\mathsf{T}(f) = O \Rightarrow f = 0$ (four zeros; Lagrange), and both dimensions are 4: an isomorphism (Theorem 2.5).

Theorem 2.19

Finite-dimensional V, W over $\mathbb{F}$ are isomorphic iff $\dim V = \dim W$.

Proof. "Only if": the Lemma. "If": with bases $\{v_i\}$, $\{w_i\}$, take linear $\mathsf{T}(v_i) = w_i$ (Theorem 2.6); $\mathrm{R}(\mathsf{T}) = \mathrm{span}\{w_i\} = W$, so T is onto, hence one-to-one (Theorem 2.5). ■

Corollary

$V \cong \mathbb{F}^n$ iff $\dim V = n$.

Proof. $\dim \mathbb{F}^n = n$; apply Theorem 2.19 (if $V \cong \mathbb{F}^n$, V is finite-dimensional by the Lemma). ■

Theorem 2.20

Let $\dim V = n$, $\dim W = m$ with ordered bases β, γ. Then $\Phi : \mathcal{L}(V, W) \to \mathrm{M}_{m\times n}(\mathbb{F})$, $\Phi(\mathsf{T}) = [\mathsf{T}]_\beta^\gamma$, is an isomorphism.

Proof. Φ is linear (Theorem 2.8); for each A there is exactly one linear T with $\mathsf{T}(v_j) = \sum_i A_{ij} w_i$, i.e. $\Phi(\mathsf{T}) = A$ (Theorem 2.6). ■

Corollary

$\dim \mathcal{L}(V, W) = mn$.

Proof. Theorems 2.20 and 2.19. ■

Definition (Standard representation)

For an ordered basis β of n-dimensional V, $\phi_\beta : V \to \mathbb{F}^n$, $\phi_\beta(x) = [x]_\beta$, is the **standard representation** of V with respect to β.

Example 6

$\beta = \{(1,0),(0,1)\}$, $\gamma = \{(1,2),(3,4)\}$, $x = (1,-2) = -5(1,2) + 2(3,4)$: $\phi_\beta(x) = (1,-2)^t$, $\phi_\gamma(x) = (-5,2)^t$.

Theorem 2.21

ϕ_β is an isomorphism.

Proof [Supplement]. $[cx+y]_\beta = c[x]_\beta + [y]_\beta$ (uniqueness, Theorem 1.8), $\phi_\beta(x) = 0 \Rightarrow x = 0$, and $\dim V = \dim \mathbb{F}^n$; apply Theorem 2.5. ■

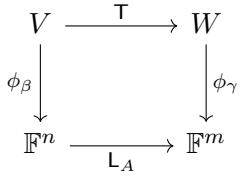


Figure. *With $A = [\mathsf{T}]^\gamma_\beta$, $\mathsf{L}_A\phi_\beta = \phi_\gamma\mathsf{T}$ (Theorem 2.14).*

Example 7

$\mathsf{T}(p) = p'$, $p = 2 + x - 3x^2 + 5x^3$: $\mathsf{L}_A\phi_\beta(p) = A(2,1,-3,5)^t = (1,-6,15)^t = \phi_\gamma(p')$.

2.5 The change of coordinate matrix

Motivation. A change of basis multiplies coordinates by Q and conjugates operator matrices by Q.

Theorem 2.22

Let β, β' be ordered bases of finite-dimensional V and $Q = [\mathsf{I}_V]^\beta_{\beta'}$. Then (a) Q is invertible; (b) $[v]_\beta = Q[v]_{\beta'}$ for all $v \in V$.

Proof.

(a) I_V is invertible, so Q is by Theorem 2.18.

(b) $[v]_\beta = [\mathsf{I}_V(v)]_\beta = [\mathsf{I}_V]^\beta_{\beta'}[v]_{\beta'}$ by Theorem 2.14. ■

Q is the **change of coordinate matrix** ($\beta' \to \beta$); column j is $[x'_j]_\beta$; Q^{-1} goes $\beta \to \beta'$.

Example 1

$\beta = \{(1,1),(1,-1)\}$, $\beta' = \{(2,4),(3,1)\}$: $(2,4) = 3(1,1) - (1,-1)$, $(3,1) = 2(1,1) + (1,-1)$, $Q = \left(\begin{smallmatrix} 3 & 2 \\ -1 & 1 \end{smallmatrix}\right)$.

A linear map $\mathsf{T} : V \to V$ is a **linear operator**.

Theorem 2.23

Let T be a linear operator on finite-dimensional V with ordered bases β, β', and $Q = [\mathsf{I}_V]^\beta_{\beta'}$. Then $[\mathsf{T}]_{\beta'} = Q^{-1}[\mathsf{T}]_\beta Q$.

Proof. $\mathsf{T} = \mathsf{IT} = \mathsf{TI}$, so by Theorem 2.11, $Q[\mathsf{T}]_{\beta'} = [\mathsf{I}]^\beta_{\beta'}[\mathsf{T}]^{\beta'}_{\beta'} = [\mathsf{T}]^\beta_{\beta'} = [\mathsf{T}]^\beta_\beta[\mathsf{I}]^\beta_{\beta'} = [\mathsf{T}]_\beta Q$. Multiply by Q^{-1}. ■

Example 2

$\mathsf{T}(a,b) = (3a-b, a+3b)$, β, β' of Example 1: $[\mathsf{T}]_\beta = \left(\begin{smallmatrix} 3 & 1 \\ -1 & 3 \end{smallmatrix}\right)$, $Q^{-1} = \frac{1}{5}\left(\begin{smallmatrix} 1 & -2 \\ 1 & 3 \end{smallmatrix}\right)$, and

$$[\mathsf{T}]_{\beta'} = Q^{-1}[\mathsf{T}]_\beta Q = \begin{pmatrix} 4 & 1 \\ -2 & 2 \end{pmatrix}.$$

Check: $\mathsf{T}(2,4) = (2,14) = 4(2,4) - 2(3,1)$.

Example 3

Reflection about $y = 2x$: $\beta' = \{(1,2), (-2,1)\}$, $\mathsf{T}(1,2) = (1,2)$, $\mathsf{T}(-2,1) = -(-2,1)$, $[\mathsf{T}]_{\beta'} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. With β standard, $Q = \left(\begin{smallmatrix} 1 & -2 \\ 2 & 1 \end{smallmatrix}\right)$ and

$$[\mathsf{T}]_\beta = Q[\mathsf{T}]_{\beta'}Q^{-1} = \frac{1}{5}\begin{pmatrix} -3 & 4 \\ 4 & 3 \end{pmatrix}, \qquad \mathsf{T}(a,b) = \tfrac{1}{5}(-3a+4b,\ 4a+3b).$$

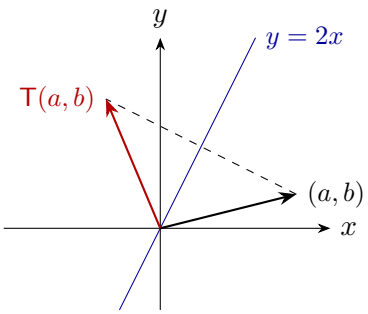


Figure. *Example 3: reflection about $y = 2x$.*

Corollary

For $A \in \mathrm{M}_{n\times n}(\mathbb{F})$ and an ordered basis γ of $\mathbb{F}^n$, $[\mathsf{L}_A]_\gamma = Q^{-1}AQ$, where the j-th column of Q is the j-th vector of γ.

Proof. Theorem 2.23 with β standard: $[\mathsf{L}_A]_\beta = A$ and $Q = [\mathsf{I}]_\gamma^\beta$ has columns the vectors of γ. ■

Example 4

For $\gamma = \{(-1,0,0), (2,1,0), (1,1,1)\}$,

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 3 \\ 0 & -1 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} -1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad [\mathsf{L}_A]_\gamma = Q^{-1}AQ = \begin{pmatrix} 0 & 2 & 8 \\ -1 & 4 & 6 \\ 0 & -1 & -1 \end{pmatrix}.$$

Definition (Similar matrices)

B is **similar** to A if $B = Q^{-1}AQ$ for some invertible Q (an equivalence relation). By Theorem 2.23, matrices of one operator are similar.

2.6 Dual spaces *

Motivation. Linear functionals form V^*, whose dual maps have transposed matrices.

A **linear functional** on V is a linear map $V \to \mathbb{F}$.

Example 1

$h(x) = \frac{1}{2\pi}\int_0^{2\pi} x(t)g(t)\,dt$ (g continuous); $g = \sin nt, \cos nt$ give Fourier coefficients.

Example 2

$\mathrm{tr} : \mathrm{M}_{n\times n}(\mathbb{F}) \to \mathbb{F}$ is a linear functional.

Example 3

The **i-th coordinate function** $f_i(x) = a_i$, $[x]_\beta = (a_1, \ldots, a_n)^t$, has $f_i(x_j) = \delta_{ij}$.

Definition (Dual space)

$V^* = \mathcal{L}(V, \mathbb{F})$. If $\dim V < \infty$, $\dim V^* = \dim V \cdot \dim \mathbb{F} = \dim V$ (Corollary to Theorem 2.20), so $V \cong V^*$. The **double dual** is $V^{**} = (V^*)^*$.

Theorem 2.24

For an ordered basis $\beta = \{x_1, \ldots, x_n\}$ with coordinate functions f_i, $\beta^* = \{f_1, \ldots, f_n\}$ is an ordered basis of V^*, and $f = \sum_{i=1}^n f(x_i)f_i$ for every $f \in V^*$.

Proof. As $\dim V^* = n$, generating suffices (Corollary 2 of Theorem 1.10). $g = \sum f(x_i)f_i$ has $g(x_j) = f(x_j)$, so $g = f$ (Corollary to Theorem 2.6). ■

β^* ($f_i(x_j) = \delta_{ij}$) is the **dual basis** of β.

Example 4

$\beta = \{(2,1),(3,1)\}$: $f_i(x_j) = \delta_{ij}$ gives $f_1(x,y) = -x + 3y$ and $f_2(x,y) = x - 2y$.

Theorem 2.25

Let V, W be finite-dimensional with ordered bases β, γ. For linear $\mathsf{T} : V \to W$, the map $\mathsf{T}^t : W^* \to V^*$, $\mathsf{T}^t(g) = g\mathsf{T}$, is linear and $[\mathsf{T}^t]_{\gamma^*}^{\beta^*} = ([\mathsf{T}]_\beta^\gamma)^t$.

Proof. $g\mathsf{T}$ is linear (Theorem 2.9), and $\mathsf{T}^t(ag + h) = (ag + h)\mathsf{T} = a\,g\mathsf{T} + h\mathsf{T}$. Let $\beta = \{x_s\}$, $\gamma = \{y_k\}$, $\beta^* = \{f_s\}$, $\gamma^* = \{g_j\}$, $A = [\mathsf{T}]_\beta^\gamma$. By Theorem 2.24, $\mathsf{T}^t(g_j) = g_j\mathsf{T} = \sum_s (g_j\mathsf{T})(x_s)f_s$, so the (i,j) entry of $[\mathsf{T}^t]_{\gamma^*}^{\beta^*}$ is

$$(g_j\mathsf{T})(x_i) = g_j\Big(\sum_k A_{ki}y_k\Big) = A_{ji}.$$

■

Example 5

$\mathsf{T}(p) = (p(0), p(2))$, $\mathsf{P}_1(\mathbb{R}) \to \mathbb{R}^2$: $[\mathsf{T}]_\beta^\gamma = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 2 \end{smallmatrix}\right)$; $\mathsf{T}^t(g_1)\colon p \mapsto p(0)$ is f_1, $\mathsf{T}^t(g_2)\colon a+bx \mapsto a+2b$ is $f_1 + 2f_2$, so $[\mathsf{T}^t] = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 2 \end{smallmatrix}\right) = ([\mathsf{T}]_\beta^\gamma)^t$.

For $x \in V$ define $\hat{x} \in V^{**}$ by $\hat{x}(f) = f(x)$; $\hat{x}$ is linear in f.

Lemma

If $\dim V < \infty$, $x \in V$ and $\hat{x}(f) = 0$ for all $f \in V^*$, then $x = 0$.

Proof. If $x \neq 0$, extend to a basis $\{x_1 = x, \ldots, x_n\}$; its dual f_1 has $f_1(x) = 1$. ■

Theorem 2.26

For finite-dimensional V, $\psi : V \to V^{**}$, $\psi(x) = \hat{x}$, is an isomorphism.

Proof. Linear: $\widehat{cx+y}(f) = c\hat{x}(f) + \hat{y}(f)$; one-to-one (Lemma); $\dim V^{**} = \dim V$; apply Theorem 2.5. ■

(ψ is basis-free: a *natural* isomorphism.)

Corollary

If V is finite-dimensional, every ordered basis of V^* is the dual basis of some basis of V.

Proof. Write the dual of a basis $\{f_j\}$ of V^* as $\{\hat{x}_i\}$ (Theorems 2.24, 2.26); $\{x_i\}$ is a basis (ψ isomorphism) and $\delta_{ij} = \hat{x}_i(f_j) = f_j(x_i)$. ■

2.7 Homogeneous linear differential equations with constant coefficients *

Motivation. Solutions of an nth-order equation form $\mathrm{N}(p(D))$, of dimension n, with a basis of exponentials.

A spring (constant k, mass m) obeys $y'' + \frac{k}{m}y = 0$.

Example 1

$y = \sin\sqrt{k/m}\,t$ is a solution; $y = t$ is not.

A **homogeneous linear differential equation with constant coefficients** is

$$a_n y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1 y' + a_0 y = 0, \qquad a_i \in \mathbb{C},\ a_n \neq 0. \tag{2.1}$$

We take $a_n = 1$ and seek solutions in $\mathcal{F}(\mathbb{R}, \mathbb{C})$.

Definition (Derivative of complex functions)

$x = x_1 + ix_2 \in \mathcal{F}(\mathbb{R}, \mathbb{C})$ is **differentiable** if x_1, x_2 are, with $x' = x_1' + ix_2'$.

Example 2

$x(t) = \cos 2t + i \sin 2t$ has $x'(t) = -2\sin 2t + 2i\cos 2t$; and $x^2 = \cos 4t + i \sin 4t$.

Theorem 2.27

Every solution of (2.1) has derivatives of all orders.

Proof. $x^{(n)} = -\sum_{i<n} a_i x^{(i)}$ is differentiable; inductively, if $x^{(k)}$ exists for $k \le m$ $(m \ge n)$, then $x^{(m)} = -\sum_{i<n} a_i x^{(i+m-n)}$ is differentiable. ■

Example 3

For $y'' + 4y = 0$, $y'' = -4y$, so $y''' = -4y'$, and so on.

Definition (C^∞; differential operators)

$C^\infty \subseteq \mathcal{F}(\mathbb{R}, \mathbb{C})$: functions with derivatives of all orders; $D(x) = x'$ is linear on it. For complex $p(t) = \sum a_i t^i$, $\deg p > 0$, $p(D) = \sum a_i D^i$ is a **differential operator** of order $\deg p$. The **auxiliary polynomial** of (2.1) is $p(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_0$, and (2.1) reads $p(D)(y) = 0$.

Theorem 2.28

The solutions of (2.1) are exactly $\mathrm{N}(p(D))$.

Proof. By Theorem 2.27 all solutions lie in C^∞, where the equation is $p(D)(y) = 0$. ■

Corollary

The solution set of (2.1) is a subspace of C^∞, the **solution space**.

Proof. Theorems 2.28 and 2.1. ■

Definition (Complex exponential)

For $c = a + ib$, $e^c = e^a(\cos b + i \sin b)$ (**Euler's formula**: $e^{ib} = \cos b + i \sin b$). Then $e^{c+d} = e^c e^d$ and $e^{-c} = 1/e^c$. $f(t) = e^{ct}$ is an **exponential function**.

Theorem 2.29

$\frac{d}{dt} e^{ct} = ce^{ct}$.

Proof [Supplement]. With $c = a + ib$, $e^{ct} = e^{at}\cos bt + ie^{at}\sin bt$, whose derivative is $e^{at}(a\cos bt - b\sin bt) + ie^{at}(a\sin bt + b\cos bt) = (a + ib)e^{at}(\cos bt + i\sin bt)$. ■

Theorem 2.30

The solution space of $y' + a_0 y = 0$ has dimension 1 and basis $\{e^{-a_0 t}\}$.

Proof. $e^{-a_0 t}$ solves it (Theorem 2.29). For a solution x, $(e^{a_0 t}x)' = e^{a_0 t}(a_0 x + x') = 0$, so $e^{a_0 t}x$ is constant (real and imaginary parts): $x = ke^{-a_0 t}$. ■

Corollary

For $c \in \mathbb{C}$, $\mathrm{N}(D - c\mathsf{I})$ has basis $\{e^{ct}\}$.

Proof. Theorem 2.30 with $a_0 = -c$. ■

Theorem 2.31

If c is a zero of the auxiliary polynomial $p(t)$, then e^{ct} is a solution.

Proof. $p(t) = q(t)(t-c)$ gives $p(D) = q(D)(D-c\mathsf{I})$ and $p(D)(e^{ct}) = q(D)((D-c\mathsf{I})e^{ct}) = q(D)(0) = 0$ by the Corollary. ■

Example 4

$y'' - 3y' + 2y = 0$: $p(t) = (t-1)(t-2)$, so e^t and e^{2t} are solutions.

Theorem 2.32

For any differential operator $p(D)$ of order n, $\mathrm{N}(p(D))$ is an n-dimensional subspace of C^∞.

Lemma 1

$D - c\mathsf{I} : C^\infty \to C^\infty$ is onto for every $c \in \mathbb{C}$.

Proof. For $v \in C^\infty$, antidifferentiate real and imaginary parts of ve^{-ct} to get $W' = ve^{-ct}$; with $u = We^{ct}$, $(D - c\mathsf{I})u = W'e^{ct} + cWe^{ct} - cWe^{ct} = v$. ■

Lemma 2

Let T, U be linear operators on V with U onto and $\mathrm{N}(\mathsf{T})$, $\mathrm{N}(\mathsf{U})$ finite-dimensional. Then $\mathrm{N}(\mathsf{TU})$ is finite-dimensional and $\dim \mathrm{N}(\mathsf{TU}) = \dim \mathrm{N}(\mathsf{T}) + \dim \mathrm{N}(\mathsf{U})$.

Proof. Take bases $\{u_1, \ldots, u_p\}$ of $\mathrm{N}(\mathsf{T})$, $\{w_1, \ldots, w_q\}$ of $\mathrm{N}(\mathsf{U})$, and v_i with $\mathsf{U}(v_i) = u_i$ (U onto). We show $\{v_1, \ldots, v_p, w_1, \ldots, w_q\}$ is a basis of $\mathrm{N}(\mathsf{TU})$ of $p + q$ distinct vectors.

Step 1 (inside). $\mathsf{TU}(v_i) = \mathsf{T}(u_i) = 0$ and $\mathsf{TU}(w_j) = \mathsf{T}(0) = 0$.

Step 2 (spans). If $\mathsf{TU}(v) = 0$, then $\mathsf{U}(v) = \sum a_i u_i = \mathsf{U}(\sum a_i v_i)$, so $v - \sum a_i v_i \in \mathrm{N}(\mathsf{U})$ equals $\sum b_j w_j$.

Step 3 (independent). Apply U to $\sum a_i v_i + \sum b_j w_j = 0$: $\sum a_i u_i = 0$, so $a_i = 0$, then $b_j = 0$ (distinctness similarly). ■

Proof of Theorem 2.32. Induction on n; $n = 1$: $p(D)$ is a nonzero multiple of $D - c\mathsf{I}$ (Theorem 2.30). For $n > 1$, $p(D) = q(D)(D - c\mathsf{I})$ with $\deg q = n - 1$; $D - c\mathsf{I}$ is onto (Lemma 1), $\dim \mathrm{N}(q(D)) = n - 1$ (induction), $\dim \mathrm{N}(D - c\mathsf{I}) = 1$; Lemma 2 gives n. ■

Corollary

The solution space of any nth-order equation (2.1) is an n-dimensional subspace of C^∞.

Proof. Theorems 2.28 and 2.32. ■

Theorem 2.33

For distinct $c_1, \dots, c_n \in \mathbb{C}$, $\{e^{c_1 t}, \dots, e^{c_n t}\}$ is linearly independent.

Proof. Induction on n. Apply $D - c_n \mathsf{I}$ to $\sum_i b_i e^{c_i t} = 0$: $\sum_{i<n} b_i (c_i - c_n) e^{c_i t} = 0$, so $b_i = 0$ for $i < n$ (induction), then $b_n = 0$. ■

Corollary

If the auxiliary polynomial of an nth-order equation (2.1) has n distinct zeros $c_1, \dots, c_n$, then $\{e^{c_1 t}, \dots, e^{c_n t}\}$ is a basis of the solution space.

Proof. They are n independent solutions (Theorems 2.31, 2.33) in an n-dimensional space. ■

Example 5

$y'' + 5y' + 4y = 0$: $p(t) = (t+1)(t+4)$, so the solutions are $b_1 e^{-t} + b_2 e^{-4t}$.

Example 6

$y'' + 9y = 0$: $p(t) = (t-3i)(t+3i)$, basis $\{e^{3it}, e^{-3it}\}$, or $\{\cos 3t, \sin 3t\}$ as $\cos 3t = \frac{1}{2}(e^{3it} + e^{-3it})$, $\sin 3t = \frac{1}{2i}(e^{3it} - e^{-3it})$.

Lemma

If the auxiliary polynomial is $(t-c)^n$, then $\beta = \{e^{ct}, te^{ct}, \dots, t^{n-1}e^{ct}\}$ is a basis of the solution space.

Proof.

Step 1 (solutions). $(D - c\mathsf{I})(t^k e^{ct}) = k t^{k-1} e^{ct}$, so $(D - c\mathsf{I})^n (t^k e^{ct}) = 0$ for $k < n$.

Step 2 (independent). $\sum_k b_k t^k e^{ct} = 0 \Rightarrow \sum b_k t^k \equiv 0 \Rightarrow b_k = 0$; with dim $= n$ (Theorem 2.32), β is a basis. ■

Example 7

$y^{(4)} - 4y^{(3)} + 6y^{(2)} - 4y' + y = 0$: $p(t) = (t-1)^4$, so the solutions are $(b_1 + b_2 t + b_3 t^2 + b_4 t^3) e^t$.

Theorem 2.34

If the auxiliary polynomial is $(t - c_1)^{n_1} \cdots (t - c_k)^{n_k}$ with distinct c_i, then $\{e^{c_1 t}, te^{c_1 t}, \dots, t^{n_1 - 1} e^{c_1 t}, \dots, e^{c_k t}, \dots, t^{n_k - 1} e^{c_k t}\}$ is a basis of the solution space.

Proof.

Step 1 (solutions). Factors of $p(D)$ commute, so each $t^j e^{c_i t}$ $(j < n_i)$, killed by $(D - c_i \mathsf{I})^{n_i}$ (Lemma), is killed by $p(D)$.

Step 2 (independent) [Supplement]. With $n = \sum n_i$ functions, independence suffices. Let $\sum_i q_i(t) e^{c_i t} = 0$, $\deg q_i < n_i$; induct on k. For $a \neq 0$, $(D - c\mathsf{I})(q(t) e^{(c+a)t}) = (q' + aq) e^{(c+a)t}$, and

$q \mapsto q' + aq$ preserves degree. Applying $(D - c_k \mathsf{I})^{n_k}$ kills the k-th term (Step 1) and maps the others to $\tilde{q}_i e^{c_i t}$, $\deg \tilde{q}_i = \deg q_i$. By induction $\tilde{q}_i = 0$, so $q_i = 0$ $(i < k)$; then $q_k e^{c_k t} = 0$ gives $q_k = 0$. ■

Example 8

$y''' - 4y'' + 5y' - 2y = 0$: $p(t) = (t-1)^2(t-2)$, so the solutions are $b_1 e^t + b_2 t e^t + b_3 e^{2t}$.